

An Iff Classification of Gaussian Power Series with the Peres–Virág Zero Process

Literature-implied resolution packet for arXiv:2103.11947

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Classification: literature-implied answer; high confidence. Sodin’s 2000 Calabi–rigidity theorem implies an exact converse to the same–Bergman–process part of Mukeru–Mulaudzi. A nondegenerate proper complex Gaussian power series has the Peres–Virág zero process if and only if it is, in law, a zero-free analytic multiple of the iid Gaussian series. When its coefficient covariance is a bounded invertible operator on ℓ^2 , this is equivalent to the inverse covariance being Toeplitz. The classification of series whose zeros form an arbitrary determinantal process remains open.

1 Source problem and exact scope

On PDF page 2 of [1], the authors formulate the broad task of determining dependent Gaussian coefficient sequences whose power-series zeros are determinantal. Their Theorem 1 on the same page proves a concrete sufficient condition: if the coefficient covariance is G^{-1} for a suitable positive invertible Toeplitz matrix G , then the zeros have exactly the Peres–Virág law, the determinantal process with Bergman kernel

$$L(z, w) = \frac{1}{\pi(1 - z\bar{w})^2}.$$

The result below gives a complete converse for this same zero process. Its core is already contained in Sodin’s Theorem 2 [2], which appears on PDF page 5 and predates the source paper by two decades.

Throughout, “proper” means zero pseudo-covariance. Nondegeneracy means that the analytic Gaussian law has no nontrivial deterministic linear relation; it is the hypothesis corresponding to Sodin’s linearly independent components.

2 Exact classification

Let

$$f(z) = \sum_{n \geq 0} \xi_n z^n, \quad g(z) = \sum_{n \geq 0} \eta_n z^n,$$

where f is a nondegenerate centred proper complex Gaussian analytic function on $\mathbb{D}$ and the η_n are iid standard proper complex Gaussians. Write

$$K_f(z, w) = \mathbb{E}[f(z)\overline{f(w)}], \quad K_0(z, w) = \frac{1}{1 - z\bar{w}}.$$

Theorem 1 (Peres–Virág iff classification). *The following are equivalent.*

1. The zero process of f has the Peres–Virág law.
2. The expected zero measure of f equals that of g , namely

$$\mathbb{E}n_f(dz) = \frac{1}{\pi(1-|z|^2)^2} dA(z).$$

3. There is a zero-free analytic function h on $\mathbb{D}$ such that

$$K_f(z, w) = \frac{h(z)\overline{h(w)}}{1 - z\overline{w}}.$$

4. As random analytic functions, $f \stackrel{d}{=} hg$.

5. If $h(z) = \sum_{j \geq 0} h_j z^j$, then the coefficient sequence has the same joint law as

$$\xi_n = \sum_{k=0}^n h_{n-k} \eta_k, \quad n \geq 0.$$

In particular, equality of the one-point intensity already forces equality of the entire zero process.

Proof. The implication (1) $\Rightarrow$ (2) is immediate. By the Edelman–Kostlan formula,

$$\mathbb{E}n_f = \frac{1}{4\pi} \Delta \log K_f(z, z) dA(z),$$

with the equivalent $\partial\bar{\partial}$ normalization. Thus (2) makes

$$\log K_f(z, z) - \log K_0(z, z)$$

harmonic. Since $\mathbb{D}$ is simply connected, it equals $\log |h(z)|^2$ for a single-valued zero-free analytic h . Hence

$$K_f(z, z) = |h(z)|^2 K_0(z, z).$$

The diagonal is a uniqueness set for kernels analytic in z and anti-analytic in w : expand at any point, or differentiate in z and $\bar{w}$. Polarization therefore gives (3). This is exactly the kernel form of Sodin’s Calabi-rigidity theorem.

A centred proper complex Gaussian analytic law is determined by its covariance kernel. The right-hand side in (3) is the covariance kernel of hg , proving (4). Comparing Taylor coefficients proves (5). Conversely, (5) gives $f \stackrel{d}{=} hg$; because h has no zeros, multiplication by h leaves every zero unchanged. Thus (5) $\Rightarrow$ (1), completing the cycle. $\square$

3 The inverse-Toeplitz converse

The preceding theorem needs no boundedness assumption on the infinite coefficient covariance. Under the standard bounded-operator hypothesis it also characterizes the source’s matrix condition.

Corollary 2 (Toeplitz inverse iff). *Assume the coefficient covariance matrix $C = (\mathbb{E}\xi_j \overline{\xi_k})$ defines a bounded positive invertible operator on $\ell^2(\mathbb{N}_0)$. Then the five conditions of Theorem 1 hold if and only if C^{-1} is a bounded positive invertible Toeplitz operator.*

Proof. Under item (5), let T_h be the lower-triangular analytic Toeplitz matrix of multiplication by h on H^2 . Then

$$C = T_h T_h^*.$$

Boundedness and invertibility of C force T_h to be bounded and bijective; equivalently $h, 1/h \in H^\infty$. Therefore

$$C^{-1} = T_{1/h}^* T_{1/h} = T_{|1/h|^2},$$

which is Toeplitz.

Conversely, write $C^{-1} = T_\varphi$. Positivity and bounded invertibility give $0 < c \leq \varphi \leq C_0$ almost everywhere. Spectral factorization yields an outer a with $\varphi = |a|^2$ and $a, 1/a \in H^\infty$. Hence

$$C^{-1} = T_a^* T_a, \quad C = T_{1/a} T_{1/a}^*.$$

Taking $h = 1/a$ gives item (3) of Theorem 1. □

Scope boundary. The source’s introductory wording asks about any determinantal zero process. The theorem above classifies exactly the process appearing in its main result: the Bergman/Peres–Virág process. For an unspecified determinantal kernel, the one-point intensity does not encode the determinantal representation, and the general Kac–Rice correlation identities lead to a substantially broader functional classification problem. No claim about that unrestricted problem is made here.

Literature status. Sodin’s Theorem 2 says that two nondegenerate Gaussian analytic functions with the same expected zero measure differ, at the analytic-vector level, by a zero-free analytic scalar and a constant unitary. Applying it to $(1, z, z^2, \dots)$ gives Theorem 1. The coefficient convolution and Corollary 2 are short operator-theoretic translations of that established theorem. The target paper does not cite Sodin. Searches on 11 August 2026 found no paper explicitly recording this converse in the target’s coefficient-matrix language, so the result is classified as a literature implication rather than a new proof claim.

Evidence locations. Source: arXiv:2103.11947, PDF page 2 (broad problem and Theorem 1), continuing on page 3. Supporting source: arXiv:math/0007030, Section 3, Theorem 2 on PDF page 5; its proof continues on pages 5–6.

References

- [1] S. Mukeru and M. P. Mulaudzi, *Zeros of Gaussian Power Series, Hardy Spaces and Determinantal Point Processes*, arXiv:2103.11947 (2021).
- [2] M. Sodin, *Zeros of Gaussian Analytic Functions*, Math. Res. Lett. **7** (2000), 371–381; arXiv:math/0007030.
- [3] Y. Peres and B. Virág, *Zeros of the I.I.D. Gaussian Power Series: A Conformally Invariant Determinantal Process*, Acta Math. **194** (2005), 1–35.